

BTM481
Special Topics in Technology Management I:
Managerial Economics of Climate & Energy
Technologies

Spring Semester of 2025

Instructor: Jiyong Eom, Ph.D.

Bld E11 #103
10:30—11:50 a.m. on Tuesday and Thursday

As of December 28, 2024

Course Overview

The course "*Managerial Economics of Climate & Energy Technologies*" examines the intersection of economics, management, and energy systems in the context of climate change. Students will gain a deep understanding of the economic principles underpinning energy markets, the dynamics of renewable and non-renewable energy technologies, and the role of policies in shaping energy transitions. By integrating theoretical frameworks with real-world case studies, this course equips students to make informed decisions in managing energy projects, analyzing market structures, and navigating the complexities of climate policies.

Course Objectives

By the end of this course, students will be able to:

- Understand and apply fundamental economic principles to analyze energy markets and technologies.
- Evaluate market structures, pricing, and supply-demand drivers in energy systems.
- Assess the economic, social, and environmental implications of energy technologies.
- Analyze the role of policies, incentives, and market-based instruments in addressing energy-related externalities.
- Develop strategic management approaches for businesses navigating the energy transition.
- Conduct investment and risk analyses for energy projects.
- Critically assess global and local climate and energy policies and their impact on markets.
- Present data-driven insights and recommendations in group projects on climate topics.

Teaching Team

Instructor

Jiyong Eom, Ph.D.

Classes: 10:30-11:50 Tuesday and Thursday

Phone: 042-350-4161

Email: eomjiyong@kaist.ac.kr

Office Hours: Thursday afternoon @ N5 2120 (to be arranged upon request)

Teaching Assistant

TBA, Master's Program Student

Prerequisite and Course Requirements

Understanding microeconomics at an undergraduate level is essential for this course. If you are not familiar with fundamental microeconomics principles, you can consider taking introductory courses first. Alternatively, you should be prepared to dedicate considerable time to study outside class. Special microeconomics tutorials or teaching assistant sessions might be offered.

Textbooks

The main textbooks for this course include (unfortunately, I could not find any single textbook covering managerial economic issues on climate and energy technologies):

- Instructor's lecture notes
- Peter M. Schwarz, *Energy Economics*, 2nd Ed., Routledge
- Charles D. Kolstad, *Intermediate Environmental Economics*, 2nd Ed., Oxford University Press
- Richard S. J. Tol, 2019, *Climate Economics: Economic Analysis of Climate, Climate Change and Climate Policy*, 3rd Ed., Edward Elgar Publishing
- *IPCC, 2023: Summary for Policymakers. In: Climate Change 2023: Synthesis Report.* Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)]. IPCC, Geneva, pp. 1-34, doi: 10.59327/IPCC/AR6-9789291691647.001, https://www.ipcc.ch/report/ar6/syr/downloads/report/IPCC_AR6_SYR_SPM.pdf
- Hal R. Varian, *Microeconomic Analysis*, 3rd Ed., Norton—Advanced microeconomics textbook with solid theoretical coverage and insights, useful for those interested in conducting mathematical modeling research in this area

Learning Framework

Class Participation & Preparation

Active participation and thorough preparation are essential components of this course. Students are required to preview lecture notes and assigned readings before each class to engage meaningfully in discussions. Lecture notes will be distributed in advance for preparation. Class participation is mandatory, and absences will result in penalties. To foster dynamic classroom dialogue, active student contributions will be credited. Additionally, brief quizzes covering previously discussed topics may be conducted at the start of select classes.

Paper Presentations

Students will work in groups of two or three to present assigned research papers, with group assignments finalized during the first week of the course. Presentations will take place during the second half of (almost) every other class. Each group is responsible for preparing and delivering a clear and comprehensive presentation of their assigned paper. Following audience feedback, including questions, comments, and suggestions, groups are required to revise and upload their presentation materials. As these materials will be used by classmates for quiz preparation, groups are expected to put their best effort into creating high-quality reviews and slides.

Paper Review Notes

All students are required to submit a one-page, single-spaced review for each research paper assigned for the week. The review should include a concise summary and a brief discussion, written in the student's own words. Students presenting a paper on a given day are exempt from submitting review notes for that paper. Hard copies of the reviews must be submitted to the instructor before the start of class.

Group Project: Research Proposal

As part of the course requirements, students will collaborate in groups to develop a research proposal that addresses a contemporary issue at the intersection of managerial economics and climate & energy technologies. This project will challenge students to apply the theoretical concepts and analytical tools learned in class to real-world problems, crafting actionable plans that offer innovative solutions or insights. The proposal should define a clear and focused research question that is academically novel, and policy- or managerially relevant. It should also include a thorough outline of relevant economic and managerial frameworks and a detailed methodology for addressing the issue. Students are encouraged to explore diverse topics, such as energy market dynamics, the economic implications of renewable energy adoption, or the role of climate policies in driving energy transitions. The final output will be a written proposal, accompanied by an in-class presentation to foster peer feedback and critical engagement.

Quizzes

Quizzes are a core part of this course, designed to reinforce key concepts and encourage consistent preparation. Administered at the start of select classes, they assess understanding of previous topics and readings. Students will also contribute quiz questions from research paper presentations, fostering peer learning and deeper engagement. This approach sharpens critical thinking and supports the course's goal of preparing students for informed decision-making in climate and energy technologies.

Evaluation Criteria

- Class Participation: 20%
- Paper Presentation: 15%
- Paper Review Notes: 15%
- Group Project: 25%
- Quizzes: 25%

Course Schedule and Assignment

Session # (Date)	Topics to cover	Paper review assignment
1 (Feb. 25)	Course introduction and objectives	
2 (Feb. 27)	Basics of Managerial Economics in the energy context	
3 (Mar. 4)	Demand, supply, and elasticity in energy markets	
4 (Mar. 6)	Marginal analysis and decision making	IPCC (2023)
5 (Mar. 11)	Market structures in energy economics	
6 (Mar. 13)	Cost structures in energy industries	Aldy et al. (2023)
7 (Mar. 18)	Overview of energy market structures	
8 (Mar. 20)	Supply-demand drivers in energy markets	Borenstein et al. (2019)
9 (Mar. 25)	Energy price formation and volatility	
10 (Mar. 27)	Environmental externalities and social costs	Flammer et al. (2021)
11 (Apr. 1)	Policy tools for addressing externalities	
12 (Apr. 3)	Comparative economics of energy technologies	Borenstein (2012)
13 (Apr. 8)	Fossil fuels vs. renewables: Transition dynamics	
14 (Apr. 10)	Technology innovation in energy	Calel et al. (2021)
15 (Apr. 15)	<i>Mid-Term Week (No Class)</i>	
16 (Apr. 17)		
17 (Apr. 22)	Emerging energy technologies	
18 (Apr. 24)	Investment decision-making in energy projects	Caldecott et al. (2021)
19 (Apr. 29)	Group project proposal presentations	
20 (May 1)	Group project proposal presentations	
21 (May 6)	<i>Substitute Holiday</i>	
22 (May 8)	Managing risks in energy investments	Joskow (2019)
23 (May 13)	Decentralized energy systems and energy storage	
24 (May 15)	Consumer behavior in energy use	Allcott (2011)
25 (May 20)	Incentives and nudging strategies	
26 (May 22)	Global climate policies	Heal (2017)

27 (May 27)	National and local energy policies	
28 (May 29)	Assessing climate risks for businesses	Nordhaus (2018)
29 (Jun. 3)	Geopolitical implications of energy transitions	
30 (Jun. 5)	Energy security and international cooperation	Li et al. (2024)
31 (Jun. 10)	<i>Presentation of Group Projects</i>	
32 (Jun. 12)		

Academic Honor Code

Academic integrity and honesty are critical values of KAIST community. It is essential to the academic integrity of this community that students do their own work and properly acknowledge the ideas, sources, and assistance upon which that work is based. As a member of KAIST BTM community, all students including those who take BTM courses are expected to adhere to the principles of truth, integrity, and respect. Failure to comply with the Honor Code may result in disciplinary action including failure of the course.

Academic dishonesty includes but is not limited to the following:

- *Cheating*: Copying from another's examination paper, solutions, or assignments or allowing another to copy from one's own.
- *Plagiarism*: Using another person's original work without giving appropriate credit to or acknowledging the authors or sources
- *Self-plagiarism*: Submitting one piece of work in more than one course without the explicit permission of the instructors involved.
- *Misrepresentation of authorship*: Submitting work as one's own which has been prepared by or purchased from another.
- *Unpermitted collaboration or aid*: Giving or receiving unpermitted aid on exams or assignments.

Any member of the BTM community who believes that violation of academic dishonesty has occurred should bring the matter to the attention of the department chair. The department chair will assign members of Academic Review Committee (학사심의회) to conduct a thorough investigation and, if necessary, request a due process to university.

Precautions

This course embraces the evolving role of technology in education and encourages the responsible use of advanced tools, including large language models (LLMs) like ChatGPT. Students are welcome to use LLMs as a resource for their assignments, provided they disclose how these tools were utilized. This disclosure must detail the specific assistance sought, such as idea generation, explanations, or support with literature surveys. The objective is to promote transparency and uphold academic integrity while leveraging these technologies to enrich the learning process. It is essential to ensure that LLMs complement, rather than replace, your original thought, critical analysis, and academic rigor.

References

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- Flammer, C., Toffel, M.W. and Viswanathan, K., 2021. Shareholder activism and firms' voluntary disclosure of climate change risks. *Strategic Management Journal*, *42*(10), pp.1850-1879.
- Heal, G., 2017. The economics of the climate. *Journal of Economic Literature*, *55*(3), pp.1046-1063.
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- Joskow, 2019. Challenges for wholesale electricity markets with intermittent renewable generation at scale: the US experience. *Oxford Review of Economic Policy*, *35*(2), pp.291-331.
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